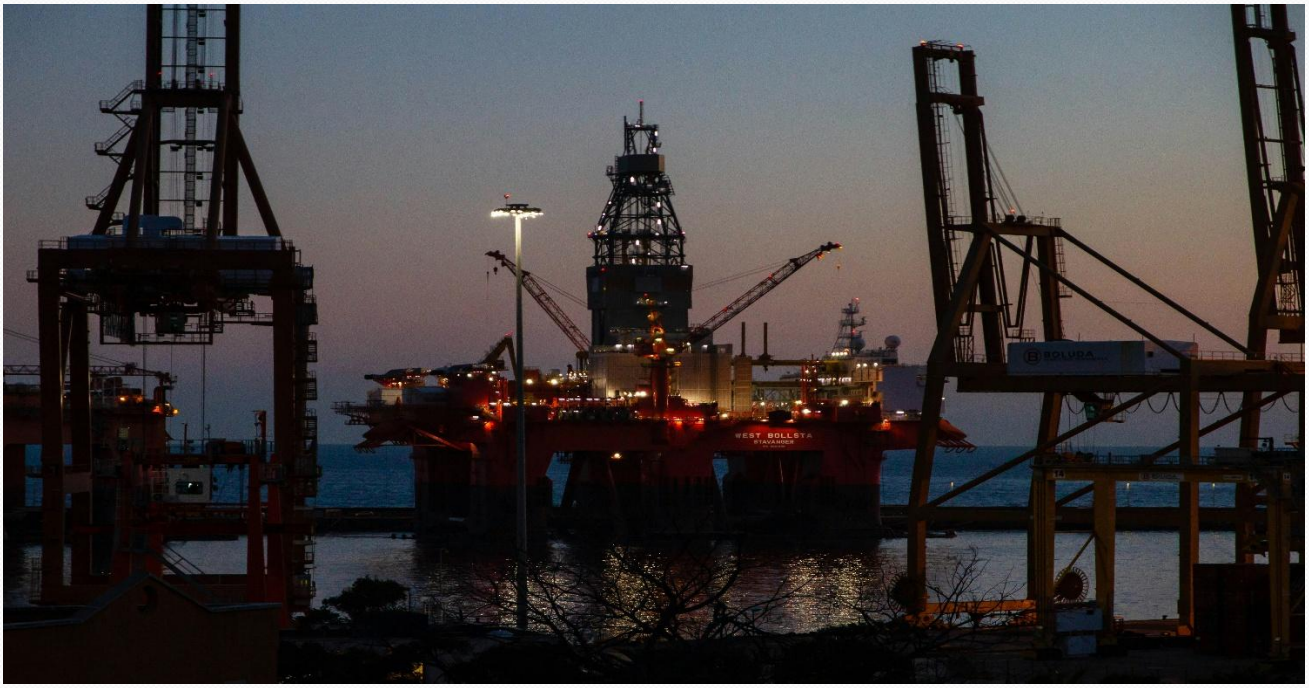


PROJECT REPORT

Volve Field: Drilling Non-Productive Time vs. Well Productivity

Python → PostgreSQL → Power BI

An end-to-end analysis of 39,395 rig hours across 26 wellbores (1992–2016)
Did the wells that were hardest to drill deliver less?
A study of non-productive time in the Volve field.

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Date: October 2026

Repository: github.com/utkarshraj94/volve-npt-analysis

Live dashboard: https://utkarshraj94.github.io/volve-npt-analysis/dashboard/

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Data: Equinor Volve open dataset (daily drilling reports and production data)

Purpose: Educational / portfolio, non-commercial

Contents

1. Executive Summary 3
2. Background and Objectives 4
2.1 Business question 4
2.2 Objectives 4
2.3 Scope 4
3. Data Sources 4
3.1 Data quality findings 5
4. Methodology 5
4.1 Project phases 5
4.2 Defining NPT 5
4.3 Classification and audit 6
4.4 Data warehouse (PostgreSQL star schema) 6
4.5 Power BI report 7
5. Results 7
5.1 Where the hours went 8
5.2 NPT Pareto: what caused it 8
5.3 NPT by wellbore 9
5.4 NPT by hole section 9
5.5 Annual trend 10
5.6 Modelled cost 10
5.7 NPT versus production 11
6. The Power BI Report 12
7. Limitations and Assumptions 15
8. Conclusions and Recommendations 16
8.1 Conclusions 16
8.2 Recommendations 16
Appendix A. Repository Structure 17
Appendix B. Key DAX Measures 18
Appendix C. How to Reproduce 19
Appendix D. Data Licence 19

1. Executive Summary

This project measures how much of the Volve drilling campaign was lost to non-productive time (NPT), where that time went, what it plausibly cost, and whether the wells that were hardest to drill were also the wells that produced the least. The work runs from raw daily drilling reports through a classified, audited dataset, a PostgreSQL star schema and a five-page Power BI report.

Headline results

Measure	Result
Logged rig time analysed	39,395 hours across 26 wellbores and 1,750 daily reports (1992–2016)
Non-productive time	7,888 hours = 20.0% of logged time
Productive time	30,510 hours = 77.4%
Rig moves / mobilisation	997 hours = 2.5% (kept separate from both)
Where the NPT went	Surface-equipment repair (36.9%) and weather (24.5%) together account for 61.4% of all NPT
Concentration	13 of 26 wellbores account for 82% of all NPT hours
Modelled cost of NPT	About $98.6M at an assumed $12,500 per rig-hour (a modelling assumption, not a reported figure)
NPT vs. production	Among the six wells with oil data, high NPT did not go with low output (small sample; see Section 7)

Main conclusions

- One in five logged rig hours was non-productive. Most of it came from two causes that are largely outside the well itself: **surface-equipment maintenance** and **weather**.
- NPT was uneven. A few short or exploratory wellbores (19-B at 78.5%, F-11 at 56.2%) had extreme rates, while 13 wellbores carried 82% of the Total lost hours.
- The data **does not support the idea that high NPT meant low production**. F-11 lost 56.2% of its rig time yet produced 7.2 million barrels, and the largest producer (F-12) ran at 17.9%. Only six wells have oil data, so this is a pattern to note, not a proof.
- The classification behind the 20.0% figure is auditable: a human reviewed 1,517 rows, and the high-confidence tier showed a 0% error rate in its audit sample.

2. Background and Objectives

2.1 Business question

Drilling is expensive, and time spent not making progress is the most visible source of avoidable cost. The Volve field (Norwegian North Sea, operated by Equinor until 2016) published its daily drilling reports and production history as an open dataset. That makes it possible to ask a concrete question: where did the downtime go, what did it plausibly cost, and did high-NPT wells deliver proportionally higher production?

2.2 Objectives

1. Extract and clean the daily drilling reports and the production history into analysis-ready tables.
2. Define NPT in a way that can be defended, classify every logged activity against it, and measure how reliable the classification is.
3. Load the result into a PostgreSQL star schema that supports repeatable analysis.
4. Quantify NPT by category, wellbore, hole section and year, and attach a modelled cost.
5. Test, descriptively, whether NPT intensity relates to production output.
6. Present the findings in an interactive Power BI report.

2.3 Scope

This is a descriptive operational review of 26 wellbores. It is not a statistical or predictive study, and cost figures are modelled. These limits are stated again, with their consequences, in Section 7.

3. Data Sources

All data comes from the Volve dataset released by Equinor ASA and the former Volve licence partners for study and research. Raw files are not committed to the repository, in line with the licence. See LICENCE-DATA.md.

Source	Content	Size and coverage
Daily drilling reports (HTML)	Operations table per day: time range, main activity, activity code, state (ok/fail), narrative, depth, hole section	1,759 files; 1,750 with an operations table; 23,447 activity rows; 26 wellbores; reports dated 1992–2016
Production data (Excel)	Daily allocated oil, gas, water and water-injection volumes, on-stream hours, pressures, well type and flow kind	15,634 daily rows; 7 wellbores; 2007-09 to 2016-12
Reference tables (created)	wellbore_map.csv (26 wellbores, parent well, production flag) and npt_taxonomy.csv (32 NPT category rows)	Versioned in the repository

3.1 Data quality findings

Production data

- No duplicate (date, wellbore) rows. 18 rows where well type and flow kind disagree; 4 negative water-volume rows.
- Date coverage per wellbore has gaps of up to 85 days (F-12 and F-14).
- Only 7 wellbores have production data, and F-4 is a water injector with no oil volume.

Drilling reports

- 9 reports have no operations table. These contribute no hours.
- 220 of 1,750 reports (12.6%) do not sum to 24 hours. Confirmed causes: rig-move handovers split one calendar day across two well files, and small 15-minute gaps from manual report compilation. Every shortfall is an exact multiple of 0.25 hours, so this is not a parsing error. NPT is therefore always summed from the logged hours, never assumed from 24 per report.
- 265 of 1,822 actual-drilling rows show depth decreasing against the previous drilling row for the same well. This was recorded as a data characteristic and does not affect the NPT hours.

4. Methodology

4.1 Project phases

Phase	Work	Main output
0	Scaffold, schema design, environment	README, requirements, .env pattern
1	File inventory of the dataset	docs/file_inventory.csv
2	Production data cleaning	data/interim/production_clean.parquet (15,634 rows)
3	HTML report parsing; phase and state extraction; NPT signal diagnostics	data/interim/drilling_activity.parquet
4	Rule-based NPT classification and 1,517-row manual review	drilling_activity_final.parquet; docs/npt_taxonomy.md
5	PostgreSQL star schema, idempotent ETL, analytical views, seven analysis queries	sql/ scripts; src/05_load_to_postgres.py
6	Power BI data model, DAX measures and five-page report	powerbi/volve_npt.pbix; volve_theme.json

4.2 Defining NPT

Phase 3 produced two candidate signals that disagreed by a factor of six: the source's own per-row state flag (state = fail), covering 3.2% of hours, and the interruption main activity, covering 21.2% of hours. Neither was used directly. The final definition is a hybrid:

1. **A row is NPT if its final Level-1 category starts with "NPT".** Classification is driven by the specific activity and its narrative, not by a single source column.
2. **Rig moves and mobilisation are a separate third category.** A rig skid between wells is neither progress nor failure, so about 997 hours are excluded from both productive and NPT.
3. **Planned and corrective work are told apart by narrative.** Circulating for a planned bottoms-up is productive; circulating to clear a pack-off is NPT. A planned geological sidetrack is productive; a sidetrack after losing the hole is NPT.
4. **The source fail flag is an override.** A row the source marked as failed is promoted to NPT even if the keyword rules called it productive.

4.3 Classification and audit

Classification ran in three re-runnable stages. A rule-based pass (src/04_classify_npt.py) applied activity-code lookups, then narrative keyword matching, then the fail override, giving each row a Level-1 and Level-2 label and a confidence of high, medium or low. A manual review then covered every low-confidence NPT-flavoured row plus fixed-size samples of the other tiers, 1,517 rows in total. A finalisation step (src/04b_apply_review.py) applied the corrections and recorded, for every row, whether a human verified it.

Confidence tier	Rows reviewed	Overridden	Error rate
High	150	0	0.0%
Medium	150	66	44.0%
Low	1,217	1,137	93.4%

Table 4.1: Manual-review error rate by confidence tier (share of reviewed rows the human overrode).

The confidence tiers behaved as intended: the rules were reliable where they said they were, and wrong where they said they were unsure. Rule labels alone gave 21.70% NPT; after manual review the figure is 20.02%.

> **Residual uncertainty**
>
> About 2,462 low-confidence productive-flavoured rows (2,180 hours) were sampled, not fully reviewed. Only 6 of 150 sampled rows (about 4%) were hidden NPT. Extrapolated, at most about 87 hours (0.2 percentage points) of NPT could remain hidden, hence the headline of 20.0% ± 0.2.

4.4 Data warehouse (PostgreSQL star schema)

The cleaned data is loaded into PostgreSQL 18 by src/05_load_to_postgres.py, which is idempotent: it drops and recreates the schema, loads the tables and rebuilds the views on every run.

Table	Rows	Purpose
fact_drilling_activity	23,447	One row per logged activity: hours, phase, NPT category, narrative, confidence, classification source
fact_daily_production	15,634	One row per well per day: oil, gas, water, injection, on-stream hours, pressures
dim_wellbore	26	Wellbore key, parent well, sidetrack, exploration and production flags
dim_npt_category	31	Level-1 / Level-2 taxonomy with is_npt and is_rig_move flags
dim_date	9,862	Calendar attributes
dim_hole_phase	10	Hole section (for example 8.5 in hole) and phase type

Table 4.2: Star schema. Both fact tables share dim_wellbore and dim_date.

Five views support the analysis: v_npt_by_wellbore, v_npt_pareto, v_npt_cost, v_wellbore_production and v_npt_vs_production. Sidetracks are separate wellbores with the parent well kept as a roll-up attribute.

4.5 Power BI report

The report imports the six tables from PostgreSQL (Import mode). Four relationships were auto-detected, all one-to-many with single-direction filtering, and dim_date connects to both fact tables, so there are no ambiguous filter paths. dim_date is marked as the date table and raw key columns are hidden from report view. All logic lives in DAX measures held in a dedicated _Measures table (Appendix B), and the measures were validated against the database views before any visual was built. A custom theme (powerbi/volve_theme.json) and a shared header with page navigation give the five pages a consistent look.

5. Results

> **How percentages are defined**
>
> Throughout this report, NPT % = NPT hours divided by total logged hours (including rig-move time in the denominator). This matches the 20.0% headline, the Power BI measure, the web dashboard and the SQL views. The views originally excluded rig moves from the denominator (20.5% field-wide); they were standardised on total hours in a later change (see Section 7).

5.1 Where the hours went

Category	Hours	Share of total
Productive	30,509.7	77.4%
NPT (all sub-categories)	7,888.3	20.0%
Rig move / mobilisation	997.0	2.5%
Total logged	**39,395.0**	**100%**

Table 5.1: Hours by top-level category.

5.2 NPT Pareto: what caused it

Level 1	Level 2	Hours	% of NPT	Cumulative
NPT - Surface Equipment	Repair/Maintenance	2,914.9	36.9%	36.9%
NPT - External	Weather	1,933.3	24.5%	61.4%
NPT - External	Labor Action	746.0	9.5%	70.9%
NPT - Downhole Equipment	Fishing	581.3	7.4%	78.3%
NPT - Downhole Equipment	Failure	426.9	5.4%	83.7%
NPT - External	Materials/Logistics	295.2	3.7%	87.4%
NPT - Other	HSE/Permits	258.1	3.3%	90.7%
NPT - Other	Operational Failure	211.0	2.7%	93.4%
NPT - Wellbore	Hole Instability	192.2	2.4%	95.8%
NPT - Wellbore	Losses	162.5	2.1%	97.9%
Other categories (4)	Well control, unclassified, waits, corrective sidetrack	167.0	2.1%	100%

Table 5.2: NPT hours by category, largest first. Cumulative column is the Pareto line.

Two categories explain over 60% of all NPT, and **five explain 83.7%.** The Labor Action category is a single documented event: an oilfield-services strike that suspended operations across 32 well-days (746 hours), which is better read as an outlier than as a recurring cause.

5.3 NPT by wellbore

Wellbore	Type	Total hrs	NPT hrs	NPT %	Note
19-B	Exploration	617	485	78.5%	Short well
F-11	Development	390	219	56.2%	Short well
19-A	Exploration	2,590	994	38.4%	Most NPT hours
F-15-C	Development	1,820	505	27.7%	
19-BT2	Exploration	1,455	395	27.1%	
F-4	Development	2,858	697	24.4%	Water injector
F-12	Development	3,764	675	17.9%	Largest producer
F-14	Development	2,946	383	13.0%	Most efficient
F-5	Development	2,171	219	10.1%	
F-1-A	Development	155	6	3.9%	Lowest; tiny well

Table 5.3: Selected wellbores, worst to best (10 of 26). Full list in the report page and database view.

- The five exploration wellbores average 29.8% NPT (2,604 of 8,733 hours) against 17.2% for the 21 development wellbores (5,285 of 30,662 hours).
- The exploration effect is not uniform: 19-B and 19-A are high, but 19-ST2 (19.3%) is about average and 19-S (14.2%) is below average.
- The two highest rates (19-B and F-11) are on short wells of under 650 hours each, so a few bad days move the percentage a lot. Absolute hours tell a different story: 19-A, F-4 and F-12 lost the most time.
- Concentration: the 13 worst wellbores by NPT hours account for 82.3% of all NPT.

5.4 NPT by hole section

Hole section	NPT hrs	Total hrs	NPT %
post-completion	2,753	8,610	32.0%
8.5 in hole	2,517	9,086	27.7%
12.25 in hole	843	4,857	17.3%
17.5 in hole	552	3,590	15.4%
26 in hole	431	2,335	18.4%
36 in hole	274	1,273	21.5%

Hole section	NPT hrs	Total hrs	NPT %
6 in hole	211	942	22.4%
plug abandon	291	4,277	6.8%

Table 5.4: NPT by hole section (sections with under 30 hours and rows with no section omitted).

The post-completion phase (workover and interventions) and the 8.5-inch section are the two heaviest in absolute hours. Plug-and-abandon is the cleanest at 6.8%. The completion phase logs 4,136 hours with no NPT assigned and is omitted from the table.

5.5 Annual trend

Year	NPT %	Year	NPT %	Year	NPT %
1992	13.3%	2008	17.5%	2014	19.5%
1993	21.0%	2009	10.7%	2015	15.8%
1997	42.7%	2010	14.8%	2016	21.0%
1998	21.9%	2012	12.4%		
2007	14.0%	2013	20.5%		

Table 5.5: NPT % by year. Years with no drilling reports (1994–96, 1999–2006, 2011) do not appear.

The 1997 exploration campaign is the outlier at 42.7%. From 2007 to 2016 the rate moved between about 11% and 21% with no clear improving trend; 2009 was the best year at 10.7%.

5.6 Modelled cost

Wellbore	NPT hours	Modelled cost	Share of total
19-A	994	$12.4M	12.6%
F-4	697	$8.7M	8.8%
F-12	675	$8.4M	8.6%
19-ST2	572	$7.1M	7.2%
F-15-C	505	$6.3M	6.4%
All 26 wellbores	**7,888**	**$98.6M**	**100%**

Table 5.6: Modelled NPT cost = NPT hours x $12,500 per hour. The rate is an assumption (Section 7).

5.7 NPT versus production

Production volumes come from the six wellbores with oil output. F-4 is a pure water injector and is excluded. F-1-C and F-5 also carry some injection rows later in life.

Wellbore	NPT %	Cum. oil (M bbl)	NPT hrs	NPT hrs per 1,000 bbl
F-12	17.9%	28.8	675	0.023
F-14	13.0%	24.8	383	0.015
F-11	56.2%	7.2	219	0.030
F-1-C	19.8%	1.1	440	0.395
F-15-D	19.3%	0.93	435	0.466
F-5	10.1%	0.24	219	0.907

Table 5.7: Oil-producing wellbores. “M bbl” here means million barrels.

- F-14 was the most efficient producer at 0.015 NPT hours per thousand barrels, with 13.0% NPT.
- F-12 produced the most oil (28.8 million barrels) at 17.9% NPT.
- F-11 lost 56.2% of its hours to NPT and still produced 7.2 million barrels.
- F-5 had low NPT (10.1%) but only 0.24 million barrels, the worst ratio. F-1-C and F-15-D are also high on this ratio. Hours per barrel is NPT hours divided by volume, so low-volume wells score badly almost by construction.

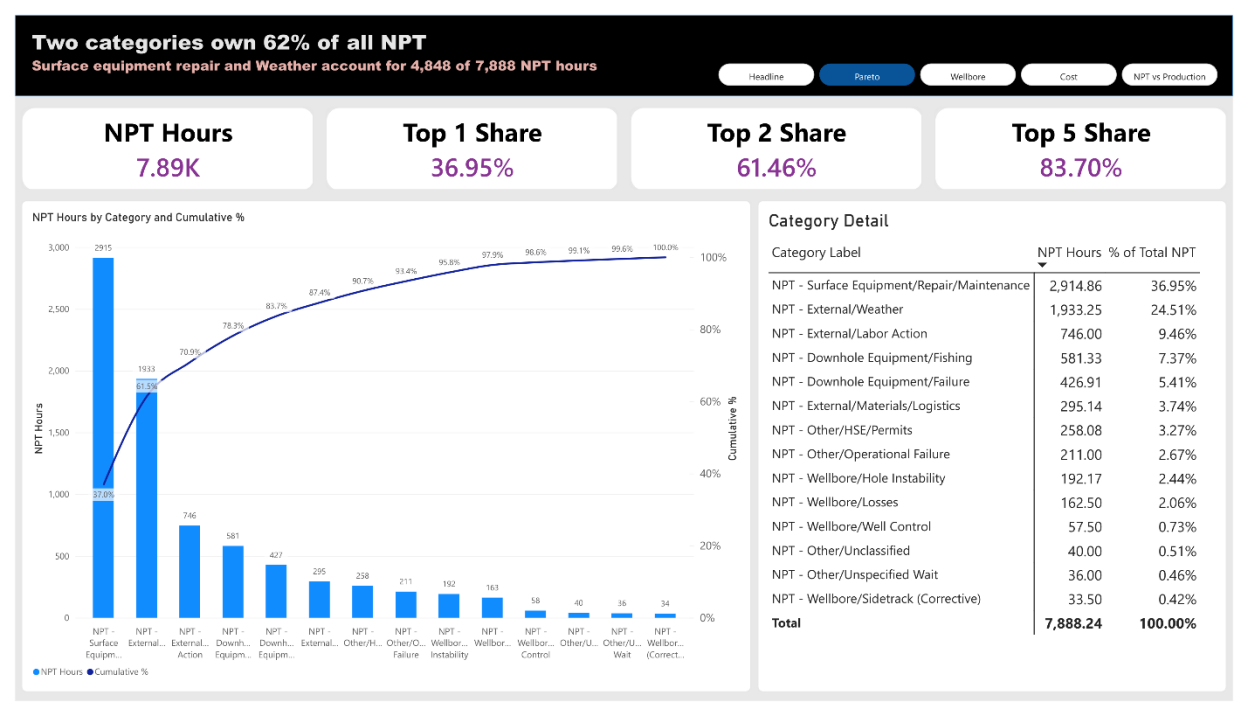
With only six wells, and F-11 the only high-NPT well among them, this supports a modest statement: in this field, NPT rate did not line up with output. It cannot show that NPT has no effect on production.

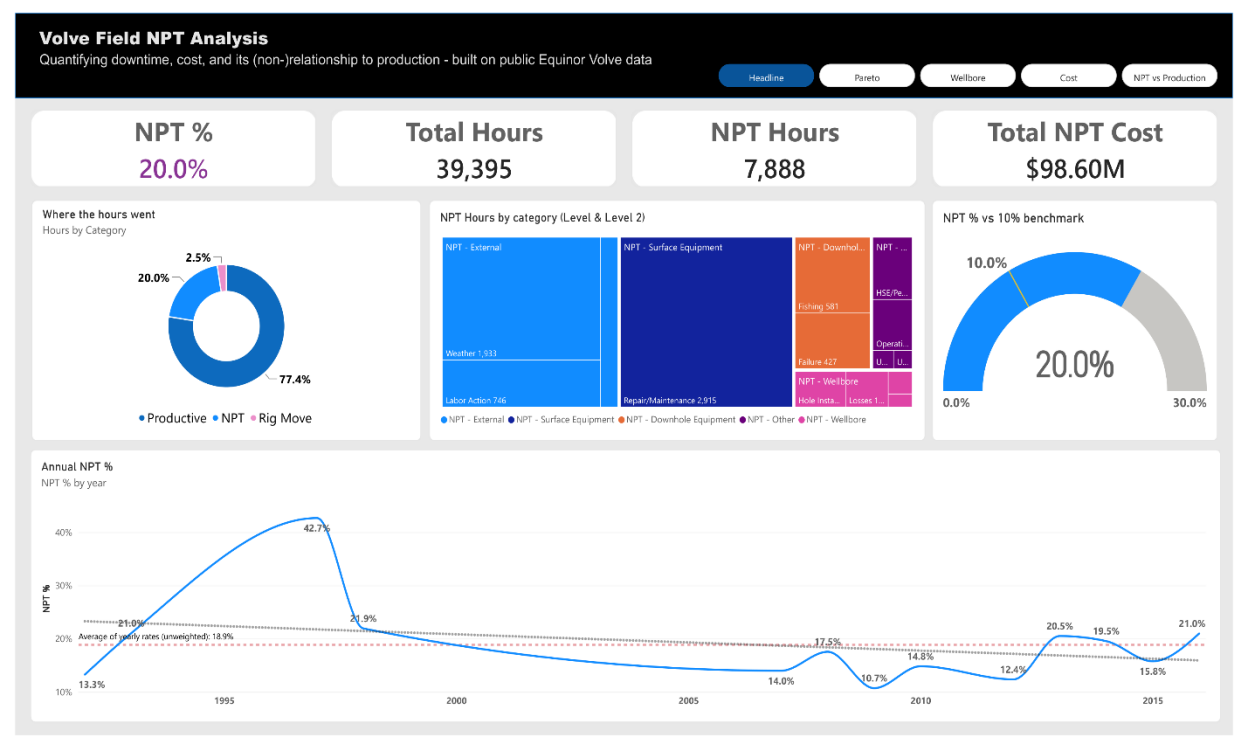
6. The Power BI Report

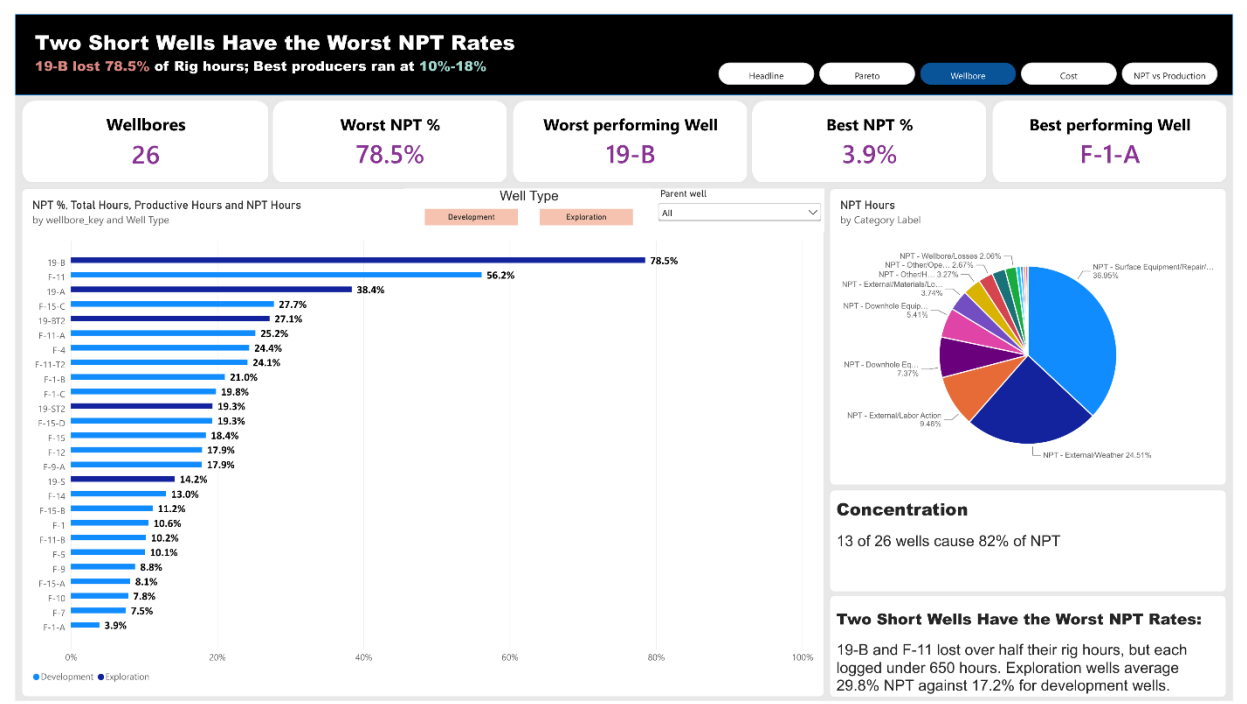
The report is built on the star schema and has five pages that share a header, a navigation bar, a blue/orange/slate palette and a rule that blue always means development and orange exploration.

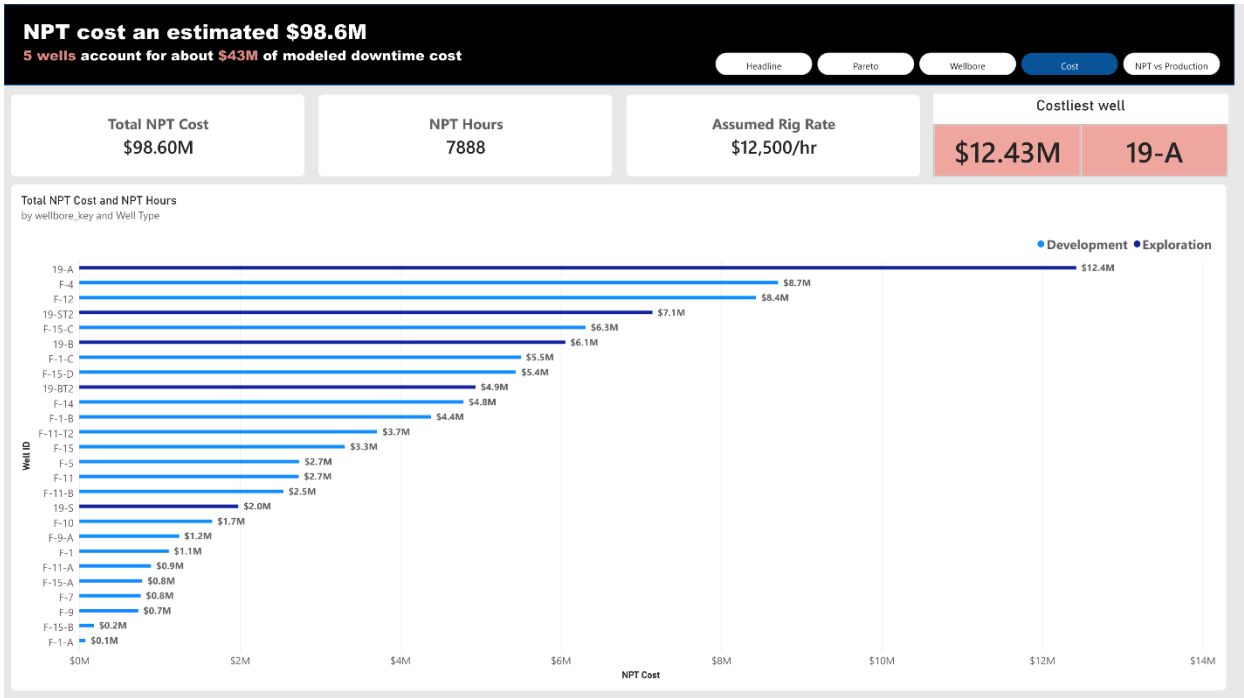
Page	Visuals	Findings
Headline	KPI cards (NPT %, total hours, NPT hours, NPT cost); hours-by-category donut with total in the centre; treemap of NPT by category; gauge against a 10% reference; annual trend line with a 20% average line	One in five rig hours was non-productive
Pareto	Column-and-line Pareto of NPT hours with cumulative %; top-1, top-2 and top-5 share cards; category table	Two categories own 62% of NPT
Wellbore	Ranked bar of NPT % by wellbore coloured by well type; parent-well and exploration slicers; worst / best NPT cards;	A few short and exploratory wells have the worst rates
Cost	Bar of modelled NPT cost by wellbore; total, rig-rate and costliest-well cards	About $98.6M of modelled cost, concentrated in a few wells
NPT vs Production	Scatter of NPT % against cumulative oil, bubble size = cost, oil producers only;	High NPT did not predict low output (small sample)

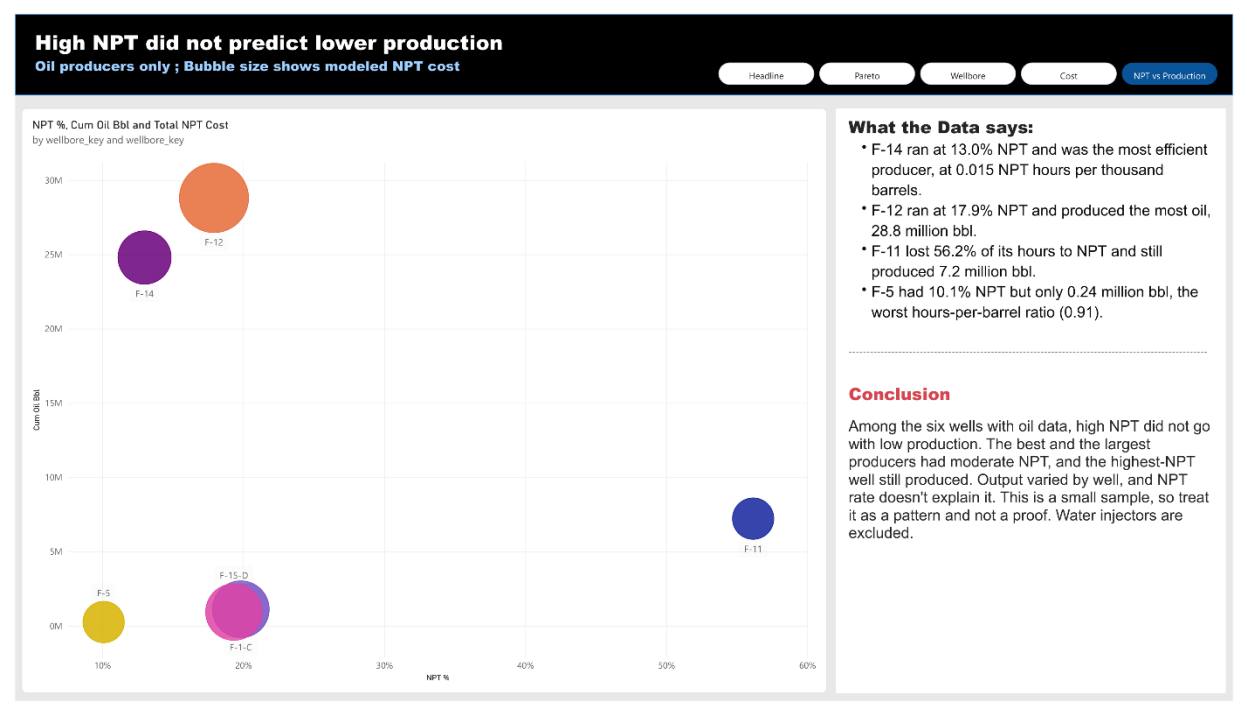
A free, interactive web version of the same five pages is hosted on GitHub Pages: **https://utkarshraj94.github.io/volve-npt-analysis/dashboard/**.


Two categories own 62% of all NPT
Surface equipment repair and Weather account for 4,848 of 7,888 NPT hours
Headline
Pareto
Wellbore
Cost
NPT vs Production
NPT Hours
7.89K
Top 1 Share
36.95%
Top 2 Share
61.46%
Top 5 Share
83.70%
NPT Hours by Category and Cumulative %
NPT Hours
Cumulative %
NPT Hours
Cumulative %
Category Detail
Category Label
NPT Hours
% of Total NPT
NPT - Surface Equipment/Repair/Maintenance 2,914.86 36.95%
NPT - External/Weather 1,933.25 24.51%
NPT - External/Labor Action 746.00 9.46%
NPT - Downhole Equipment/Fishing 581.33 7.37%
NPT - Downhole Equipment/Failure 426.91 5.41%
NPT - External/Materials/Logistics 295.14 3.74%
NPT - Other/HSE/Permits 258.08 3.27%
NPT - Other/Operational Failure 211.00 2.67%
NPT - Wellbore/Hole Instability 192.17 2.44%
NPT - Wellbore/Losses 162.50 2.06%
NPT - Wellbore/Well Control 57.50 0.73%
NPT - Other/Unclassified 40.00 0.51%
NPT - Other/Unspecified Wait 36.00 0.46%
NPT - Wellbore/Sidetrack (Corrective) 33.50 0.42%
Total 7,888.24 100.00%


Volve Field NPT Analysis
Quantifying downtime, cost, and its (non-)relationship to production - built on public Equinor Volve data
Headline
Pareto
Wellbore
Cost
NPT vs Production
NPT %
20.0%
Total Hours
39,395
NPT Hours
7,888
Total NPT Cost
$98.60M
Where the hours went
Hours by Category
Productive
NPT
Rig Move
NPT Hours by category (Level & Level 2)
NPT % vs 10% benchmark
Annual NPT %
NPT % by year
NPT %


Two Short Wells Have the Worst NPT Rates
19-B lost 78.5% of Rig hours; Best producers ran at 10%-18%
Headline
Pareto
Wellbore
Cost
NPT vs Production
Wellbores
26
Worst NPT %
78.5%
Worst performing Well
19-B
Best NPT %
3.9%
Best performing Well
F-1-A
NPT %, Total Hours, Productive Hours and NPT Hours
by wellbore_key and Well Type
Well Type
Development
Exploration
Parent well
All
19-B 78.5%
F-11 56.2%
19-A 38.4%
F-15-C 27.7%
19-BT2 27.1%
F-11-A 25.2%
F-4 24.4%
F-11-T2 24.1%
F-1-B 21.0%
F-1-C 19.8%
19-ST2 19.3%
F-15-D 19.3%
F-15 18.4%
F-12 17.9%
F-9-A 17.9%
19-S 14.2%
F-14 13.0%
F-15-B 11.2%
F-1 10.6%
F-11-B 10.2%
F-5 10.1%
F-9 8.8%
F-15-A 8.1%
F-10 7.8%
F-7 7.5%
F-1-A 3.9%
0%
20%
40%
60%
80%
100%
Development Exploration
NPT Hours
by Category Label
NPT - Surface Equipment/Repair/... 36.95%
NPT - External/Weather 24.51%
NPT - External/Labor Action 9.48%
NPT - Downhole Eq... 7.37%
NPT - Downhole Equip... 5.41%
NPT - External/Materials/Lo... 3.74%
NPT - Other/H... 3.27%
NPT - Other/Ope... 2.67%
NPT - Wellbore/Losses 2.06%
Concentration
13 of 26 wells cause 82% of NPT
Two Short Wells Have the Worst NPT Rates:
19-B and F-11 lost over half their rig hours, but each logged under 650 hours. Exploration wells average 29.8% NPT against 17.2% for development wells.


NPT cost an estimated $98.6M
5 wells account for about $43M of modeled downtime cost
Headline
Pareto
Wellbore
Cost
NPT vs Production
Total NPT Cost
$98.60M
NPT Hours
7888
Assumed Rig Rate
$12,500/hr
Costliest well
$12.43M
19-A
Total NPT Cost and NPT Hours
by wellbore_key and Well Type
Development
Exploration
Well ID
19-A $12.4M
F-4 $8.7M
F-12 $8.4M
19-ST2 $7.1M
F-15-C $6.3M
19-B $6.1M
F-1-C $5.5M
F-15-D $5.4M
19-BT2 $4.9M
F-14 $4.8M
F-1-B $4.4M
F-11-T2 $3.7M
F-15 $3.3M
F-5 $2.7M
F-11 $2.7M
F-11-B $2.5M
19-S $2.0M
F-10 $1.7M
F-9-A $1.2M
F-1 $1.1M
F-11-A $0.9M
F-15-A $0.8M
F-7 $0.8M
F-9 $0.7M
F-15-B $0.2M
F-1-A $0.1M
$0M
$2M
$4M
$6M
$8M
$10M
$12M
$14M
NPT Cost



7. Limitations and Assumptions

1. **The cost rate is an assumption.** The $12,500 per hour used for every NPT hour is hard-coded in the SQL (v_npt_cost) and in the Power BI measure. No source for it is recorded in the repository, and the README describes a different method (published North Sea day rates with a spread multiplier) that the code does not implement. Treat the dollar figures as indicative; the ranking of categories and wells is the durable result.
2. **NPT % definition was standardised late.** NPT % is NPT hours divided by total logged hours everywhere now (20.0% field-wide). Earlier, the SQL view v_npt_by_wellbore divided by hours excluding rig moves (20.5%), which is why figures quoted before the change (for example 19-A at 41.0% and F-14 at 13.3%) are slightly higher than those in this report (38.4% and 13.0%). The views, queries and dashboard export were updated to match, and the value was checked after the database was rebuilt.
3. **Small production sample.** Six wells have oil data. The NPT-versus-production comparison is descriptive and cannot support a statistical claim or any statement about cause.
4. **Classification involves judgement.** Distinguishing planned from corrective work depends on narrative text. The audit quantifies the uncertainty (about ±0.2 percentage points on the headline) but cannot remove it. The manual review was done by one reviewer.
5. **Uneven time coverage.** Drilling reports cover 1992–93, 1997–98 and 2007–2016 with gaps. Production data covers only 2007–2016 and 7 wellbores. Year-over-year comparisons mix very different campaigns.
6. **Percentages on short wells are volatile.** Wells such as 19-B and F-11 have under 650 hours, so their rates swing widely. Read them together with absolute hours.

7. **Well identity is taken from file names.** Rig-move days can be split across two well files, so a few hours may sit on the neighbouring wellbore.
8. **Production allocation.** Volumes are the dataset's allocated figures. Gaps of up to 85 days exist on two wells and were not interpolated.

8. Conclusions and Recommendations

8.1 Conclusions

- About 20% of logged rig time was non-productive, roughly 7,900 hours.
- Most NPT had causes that sit largely outside the well: equipment maintenance and weather explain 61% of it.
- NPT was concentrated: 13 wells carry 82% of the hours, and the extreme percentages belong to a few short or exploratory wellbores.
- In this field and this small sample, NPT rate did not track production output.
- The 20.0% figure rests on an audited classification, with a stated error rate and a stated residual uncertainty.

8.2 Recommendations

These follow from the data but are hypotheses for an operator to test, because the study is descriptive.

1. **Target surface-equipment reliability first.** It is the largest single cause (36.9%). A preventive-maintenance review of rig and surface equipment has the highest potential payback.
2. **Plan for weather.** At 24.5% of NPT, schedule weather-sensitive operations with contingency time and review seasonal planning.
3. **Treat the strike as an outlier.** The 746 labour-action hours are one event. Remove them when benchmarking normal performance, and manage the contract risk separately.
4. **Review exploration and sidetrack planning.** Exploration wellbores ran at 29.8% against 17.2%. The extra time is partly expected in unknown ground, but 19-A alone cost about $12.4M of modelled NPT.
5. **Judge wells by hours and output together.** NPT percentage alone misleads on short wells; pair it with absolute hours and production volume.

Appendix A. Repository Structure

Path	Contents
src/	00_download_selected.py, 01_inventory.py, 02_parse_production.py, 03_parse_drilling_reports.py, 04_classify_npt.py, 04b_apply_review.py, 05_load_to_postgres.py, build_wellbore_map.py
sql/	01_ddl_dimensions.sql, 02_ddl_facts.sql, 03_transforms.sql (views), 04_analysis_queries.sql
data/reference/	wellbore_map.csv, npt_taxonomy.csv (versioned)
data/raw, data/interim	Local only (licence); parquet intermediates
docs/	npt_taxonomy.md, quality reports, manual-review CSV, this report
powerbi/	volve_npt.pbix and volve_theme.json
run_analysis.py	Runs the analysis queries against the database

Appendix B. Key DAX Measures

All measures live in the _Measures table. Table and column names are as in the model.

```
Total Hours      = SUM(fact_drilling_activity[hours])
NPT Hours        = CALCULATE([Total Hours], fact_drilling_activity[is_npt_final] = TRUE)
Rig Move Hours   = CALCULATE([Total Hours], dim_npt_category[is_rig_move] = TRUE)
Billable Hours   = CALCULATE([Total Hours], dim_npt_category[is_rig_move] = FALSE)
Productive Hours = [Billable Hours] - [NPT Hours]
NPT %            = DIVIDE([NPT Hours], [Total Hours])
Assumed Rig Rate = 12500
NPT Cost         = [NPT Hours] * [Assumed Rig Rate]
Cum Oil Bbl      = SUM(fact_daily_production[oil_bbl])
NPT Hrs per kbbl = DIVIDE([NPT Hours], [Cum Oil Bbl] / 1000)


% of Total NPT   = DIVIDE([NPT Hours],
                     CALCULATE([NPT Hours], ALL(dim_npt_category)))


Cumulative % =
  VAR cur = [NPT Hours]
  VAR tbl = ALLSELECTED(dim_npt_category[Category Label])
  RETURN DIVIDE(
    SUMX(FILTER(tbl, [NPT Hours] >= cur), [NPT Hours]),
    SUMX(tbl, [NPT Hours]))


Top 2 Share =
  DIVIDE(SUMX(TOPN(2, ALLSELECTED(dim_npt_category[Category Label]), [NPT Hours]), [NPT
Hours]),
         CALCULATE([NPT Hours], ALLSELECTED(dim_npt_category)))


Worst NPT %  = MAXX(ALLSELECTED(dim_wellbore[wellbore_key]), [NPT %])
Best NPT %   = MINX(ALLSELECTED(dim_wellbore[wellbore_key]), [NPT %])
Worst Well   = CONCATENATEX(TOPN(1, ALLSELECTED(dim_wellbore[wellbore_key]), [NPT %], DESC),
                           dim_wellbore[wellbore_key], ", ")


Calculated columns:
Category Label = dim_npt_category[level1] & " / " & dim_npt_category[level2]
Well Type      = IF(dim_wellbore[is_exploration], "Exploration", "Development")
```

Appendix C. How to Reproduce

1. Create the virtual environment, install requirements.txt and copy .env.example to .env with the database settings. Percent-encode the password with quote_plus because it can contain special characters.
2. Obtain the Volve data under its licence and place it in data/raw (not committed).
3. Run the numbered scripts in src/ in order (00 to 04b) to produce data/interim/drilling_activity_final.parquet.
4. Run src/05_load_to_postgres.py to build the schema, load the tables and create the views.
5. Open powerbi/volve_npt.pbix in Power BI Desktop. To refresh, point the PostgreSQL connection at your own server (localhost:5432, database volve_npt).

Appendix D. Data Licence

The Volve dataset is released by Equinor ASA and the former Volve licence partners (ExxonMobil E&P Norway AS, Bayerngas Norge AS) “as is”, for study, research and development. This project is for educational and portfolio use and is non-commercial. Raw data is not redistributed. Check the original licence document that accompanies the dataset before publishing anything derived from it, including a public Power BI report, which embeds the data.